

Brainstorming & Ideation Phase

Brainstorming - Idea Generation - Prioritization

shopEZ E-commerce Marketplace | MERN Stack | June 2026

1. Problem Statement Identification

Small and medium-sized retail businesses struggle to set up and manage an online storefront efficiently due to complex setups, high subscription fees, and lack of customization. Customers demand a fast, frictionless, and secure online shopping experience with persistent carts, advanced filters, transparent pricing, and trustable payments.

PS-1: The Online Shopper (Convenience & Trust)

- **User Profile:** A busy online consumer seeking smooth retail experience across mobile and desktop.
- **Objective:** Find and purchase items quickly with minimal friction.
- **Obstacles:** Cart items disappearing on session reload, slow page loads, complex checkout pipelines, and lack of real-time inventory checks.
- **Negative Consequences:** Friction triggers frustration, resulting in high cart abandonment rates.

PS-2: The Store Admin / Seller (Operations & Control)

- **User Profile:** Business owner or manager handling catalog and orders.
- **Objective:** Manage product stocks, process orders, review customer feedback, and verify sales growth metrics.
- **Obstacles:** Clunky database updates, difficulty uploading assets, and manual order status tracking.
- **Negative Consequences:** Inventory desyncs and delayed fulfillments.

2. Brainstorming and Idea Listing

The development team collaborated to list potential solutions and architectural requirements:

- **Custom-built MERN Stack Application:** Build a fully decoupled stack (React, Node, Express, MongoDB) for absolute control over UI, performance, and backend logic.
- **JWT-based Stateless Authentication:** Implement JSON Web Tokens stored in HTTP-Only cookies to protect against XSS and maintain secure sessions.
- **Role-Based Access Control (RBAC):** Build distinct user hierarchies (Customer vs. Seller/Admin) into API and UI routers.
- **Persistent Shopping Cart:** Store cart states in React state (syncd to localized context) and associate order logs in MongoDB.
- **Interactive Admin Ledger:** Create visual charts (Chart.js) to display monthly sales, revenue progress, and real-time low-stock alarms.
- **Local Multer Image Storage:** Multer disk storage for product image uploads with static file serving.
- **Mock Stripe Payment Intents:** Simulated payment endpoint returning client secrets to model industry-standard gateway checkout.

3. Categorization and Grouping of Ideas

Category	Ideas
Core Architecture and Security	Full-stack MERN decoupling; JWT token-based authentication; Middleware enforcing access levels (user, seller, admin)
Customer Experience (CX)	Auto-saving persistent shopping cart; Catalog search with autocomplete suggestions; Multi-step checkout with coupon deductions; Verified purchase review badge and upvoting
Administrative Operations	Centralized dashboard with sales stats; Instant product CRUD operations; Customer status controllers (block/unblock); Active coupon rules compiler

4. Priority Matrix (MoSCoW Method)

Must Have (High Priority)	Should Have (Medium Priority)	Could Have (Low Priority)	Won't Have (Deferred)
JWT Authentication and RBAC	Admin Dashboard Sales Charts	Related Product Suggestions	Multi-vendor commission splits

Product Catalog Listing	Coupon Code Validation Engine	Wishlist One-Tap Move to Cart	Advanced AI recommendations
Persistent Shopping Cart	Customer Review Upvotes	Shipping Speed Selectors	Multi-currency conversions
Multi-Step Checkout Flow	Low Stock Notification Alerts	Address Book Managers	Native mobile applications
Mock Stripe Payments	Verified Buyer Review Badge	Account Info Editor	Live chat customer support